



CGRH System — HCI Analysis

From User Research to Usability Testing

UADY · FACULTY OF MATHEMATICS · HCI 2026

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Project Context



What is our app?

A web-based self-service platform for **~8,000 UADY employees**, providing access to payroll (CFDI) and various savings funds available to the university's employees.

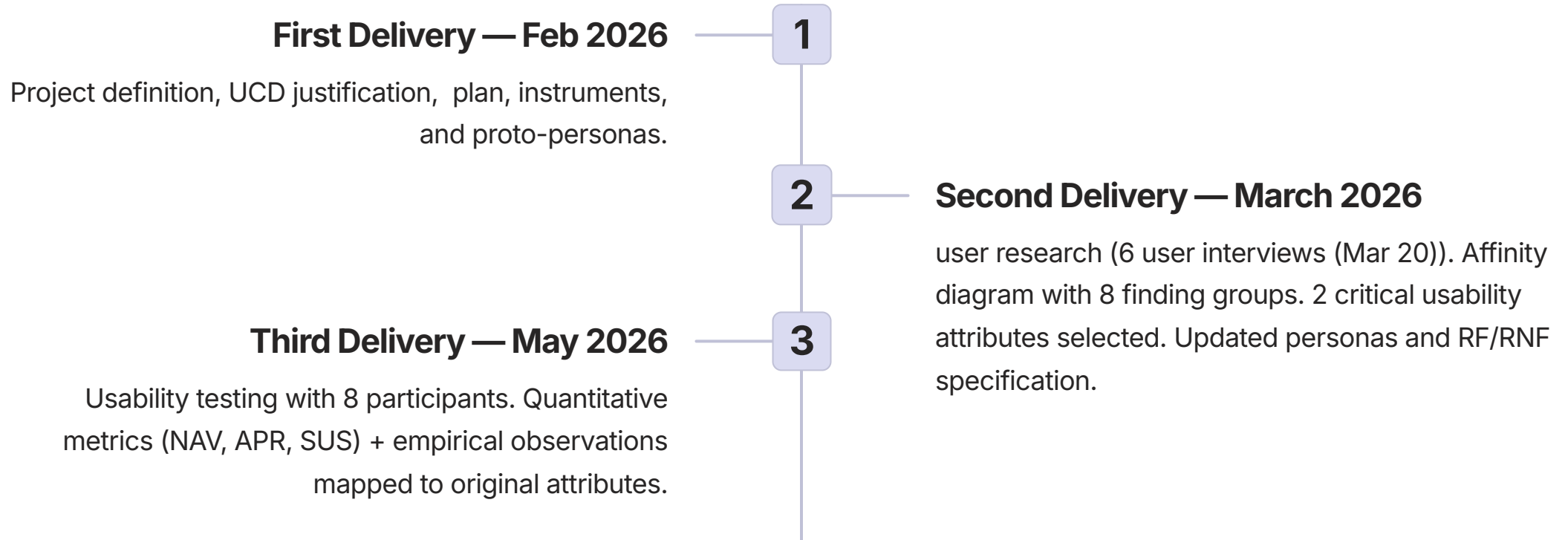
HCI Focus

Manual and Admin employees of the university, given their technology proficiency and availability.

Methodology

User-Centered Design (ISO 9241-210:2019) across 4 phases, one per delivery.

Project Evolution: Three Deliveries





Initial Research Findings

Navigation Difficulty

5/6 users said complex navigation leads to task abandonment, so it became the #1 pain point we wanted to tackle.

Low Error Autonomy

Most users stop, ask for help, or abandon when something fails instead of fixing it themselves.

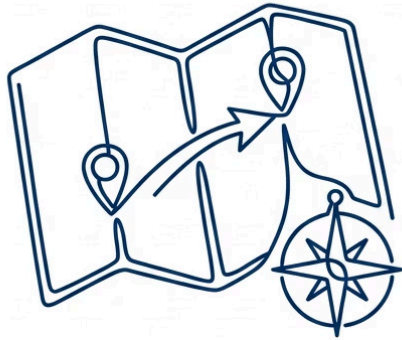
Feedback Dependence

All 6 users need explicit confirmation that an action succeeded for better clarity.

Infrequent Use

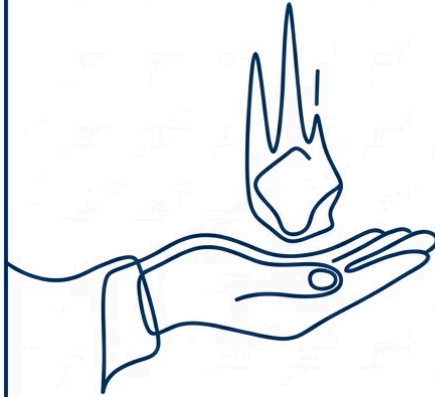
Payroll checked biweekly, savings every 6 months, users could feel like first-timers each return, especially if they are cognitively loaded already or have deficient memory due to numerous factors (i.e. age).

2 Critical Usability Attributes



1) INTUITIVE NAVIGATION

**Works instantly.
Users easily
remember path,
even after
months away.**



2) ERROR PREVENTION

**Proactively
prevents
mistakes. Essential
as users cannot
self-recover.**

Why These Two?

Derived directly from 6 user interviews and affinity diagram analysis. These attributes address the most critical failure modes for the target user group: employees with infrequent system access and limited technological abilities.

i Both attributes were later validated quantitatively in usability testing.

Usability Testing: Overall Results

8 participants (U1–U8), split between Manual and Administrative employees

6 tasks: Register, Login, Download most recent CFDI, Download all CFDI of 2024, Download Savings Fund report, Logout

93.75%

Task Success Rate

45 of 48 tasks completed successfully.

93.3

SUS Average

Rated **Excellent** (Bangor et al.). All 7 completers scored ≥ 68 .

78

Lowest SUS (U8)

Still rated **Good**. U4 left early — 3/6 tasks, SUS not completed.

Navigability & Learnability Metrics

Navigability (NAV03)

Reference optimal path: **25 clicks**. Group average: **45.75 clicks** — 83% above optimal.

- U6: 27 clicks, 0 errors — nearest to optimal. Used Tab key for navigation
- U3: 71 clicks, 1 error — completed all tasks via strong recovery
- U4: 67 clicks, 4 errors — only 3/6 tasks completed.

Perceived orientation: Very Good for 5/8 users; Medium for U3 and U4.

Learnability (APR01 — T1 Registration)

Most users: **1–10 seconds** to first correct action. U4: 40s · U3: 172s.

15 total external help events — 8 of 15 in T1 (Register) alone. U4 had 6 help events across multiple tasks.

Qualitative Scores (1–5 scale, 7 users)

- Learning Ease: 4.43
- Consistency: 4.86
- Retention: 4.71

Specific Task Findings

T1 — Registration (avg 240s, highest time & variance)

Users confused required account with Servicio Médico credentials. Password requirements not shown lead to failed attempts and help requests.

T4 — Download All CFDI 2024 (range 16s–74s)

"Select all" header checkbox not recognized, users selected records individually, inflating their click count. Affordance does not communicate purpose.

T6 — Logout (U3: 40s, 39s to first correct action)

Logout option not immediately discoverable. Placement and visibility insufficient for target user profile.

Attribute Validation & Recommendations

Attributes were confirmed

Intuitive Navigation: NAV03 avg 83% above optimal; T6 logout discoverability; registration excess steps.

Error Prevention: 8/15 help events in T1; U4 unable to self-recover; password requirements absent; select-all affordance failure.

3 Targeted Fixes

1

T1: Account label + inline password requirements

2

T4: Add "Select all" text label to header checkbox

3

T6: Improve logout visibility and placement

Requirement Change: Migration to Microsoft 365 Institutional Login

Following a client demo, the registration flow has been redesigned to use Microsoft 365 institutional SSO (Single Sign In), the same authentication system used across multiple UADY platforms, instead of independent credentials that were already validated.

HCI Impact — Positive

- Eliminates Register friction: The most problematic task (avg 240s, 8/15 help events) is resolved by the SSO.
- Structural resolution: Account type confusion and absent password requirements are solved fundamentally.
- Immediate transfer of learning: Users familiar with M365 from other institutional systems.
- Reduced credential burden: Directly addresses low digital literacy by removing an independent credential.
- Reduced scope: Password recovery and account verification shifts to UADY IT/365, simplifying the system.

HCI Impact — Considerations

- External control: M365 login screen is outside CGRH design control; error messages may or may not lack context. This is already an issue on existing platforms.
- Clear entry point: The app entry must clearly communicate "sign in with your institutional account" to prevent new confusion.
- External blocker: Users unable to log into M365 are blocked outside the app's scope, requiring UADY IT support.

Net assessment: This change resolves the most significant usability problem identified in testing and aligns with users' existing mental models. Tradeoffs are manageable, primarily ensuring clear communication at the CGRH entry point.

Conclusion

Strong Baseline

93.75% success rate and SUS 93.3 (Excellent). Problems are concentrated — not systemic.

Attributes Validated

Intuitive Navigation and Error Prevention confirmed by both quantitative and qualitative data.

Traceable Process

Three deliveries produced actionable findings from user research through testing, per ISO 9241-210:2019.

